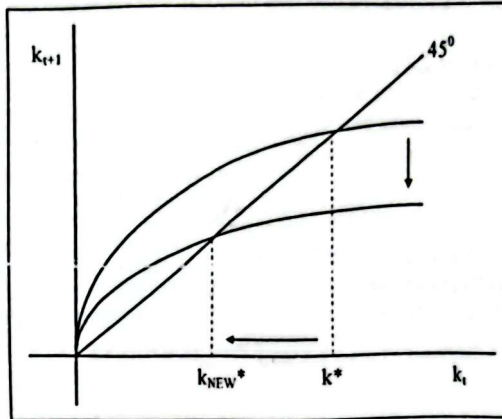


Problem 2.14

Equation (2.60) in the text describes the relationship between k_{t+1} and k_t in the special case of logarithmic utility and Cobb-Douglas production:

$$(2.60) \quad k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} (1-\alpha) k_t^\alpha.$$

(a) A rise in n shifts the k_{t+1} function down. From equation (2.60), a higher n means a smaller k_{t+1} for a given k_t . Since the fraction of their labor income that the young save does not depend on n , a given amount of capital per unit of effective labor and thus output per unit of effective labor in time t yields the same amount of saving in period t . Thus it yields the same amount of capital in period $t+1$. However, the number of individuals increases more from period t to period $t+1$ than it used to. So that capital is spread out among more individuals than it would have been in the absence of the increase in population growth and thus capital per unit of effective labor in period $t+1$ is lower for a given k_t .



(b) With the parameter B added to the Cobb-Douglas production function, $f(k) = Bk^\alpha$, equation (2.60) becomes

$$(1) \quad k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} (1-\alpha) B k_t^\alpha.$$

This fall in B causes the k_{t+1} function to shift down. See the figure from part (a). A lower B means that a given amount of capital per unit of effective labor in period t now produces less output per unit of effective labor in period t . Since the fraction of their labor income that the young save does not depend on B , this leads to less total saving and a lower capital stock per unit of effective labor in period $t+1$ for a given k_t .

(c) We need to determine the effect on k_{t+1} for a given k_t , of a change in α . From equation (2.60):

$$(2) \quad \frac{\partial k_{t+1}}{\partial \alpha} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} \left[-k_t^\alpha + (1-\alpha) \frac{\partial k_t^\alpha}{\partial \alpha} \right].$$

We need to determine $\partial k_t^\alpha / \partial \alpha$. Define $f(\alpha) \equiv k_t^\alpha$ and note that $\ln f(\alpha) = \alpha \ln k_t$. Thus

$$(3) \quad \partial \ln f(\alpha) / \partial \alpha = \ln k_t.$$

Now note that we can write

$$(4) \quad \frac{\partial f(\alpha)}{\partial \alpha} = \frac{\partial f(\alpha)}{\partial \ln f(\alpha)} \frac{\partial \ln f(\alpha)}{\partial \alpha} = \frac{1}{[\partial \ln f(\alpha) / \partial f(\alpha)]} \frac{\partial \ln f(\alpha)}{\partial \alpha},$$

and thus finally

$$(5) \quad \partial f(\alpha) / \partial \alpha = f(\alpha) \ln k_t.$$

Therefore, we have $\partial k_t^\alpha / \partial \alpha = k_t^\alpha \ln k_t$. Substituting this fact into equation (2) yields

$$(6) \quad \frac{\partial k_{t+1}}{\partial \alpha} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} \left[-k_t^\alpha + (1-\alpha) k_t^\alpha \ln k_t \right],$$

or simply

$$(7) \frac{\partial k_{t+1}}{\partial \alpha} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} \left\{ k_t^\alpha [(1-\alpha) \ln k_t - 1] \right\}.$$

Thus, for $(1-\alpha) \ln k_t - 1 > 0$, or $\ln k_t > 1/(1-\alpha)$, an increase in α means a higher k_{t+1} for a given k_t and thus the k_{t+1} function shifts up over this range of k_t 's. However, for $\ln k_t < 1/(1-\alpha)$, an increase in α means a lower k_{t+1} for a given k_t . Thus the k_{t+1} function shifts down over this range of k_t 's. Finally, right at $\ln k_t = 1/(1-\alpha)$, the old and new k_{t+1} functions intersect.

Problem 2.15

(a) We need to find an expression for k_{t+1} as a function of k_t . Next period's capital stock is equal to this period's capital stock, plus any investment done this period, less any depreciation that occurs. Thus

$$(1) K_{t+1} = K_t + sY_t - \delta K_t.$$

To convert this into units of effective labor, divide both sides of equation (1) by $A_{t+1}L_{t+1}$:

$$(2) \frac{K_{t+1}}{A_{t+1}L_{t+1}} = \frac{K_t(1-\delta) + sY_t}{A_{t+1}L_{t+1}} = \frac{K_t(1-\delta) + sY_t}{(1+n)(1+g)A_tL_t} = \frac{k_t(1-\delta) + sf(k_t)}{(1+n)(1+g)},$$

which simplifies to

$$(3) k_{t+1} = \left[\frac{1-\delta}{(1+n)(1+g)} \right] k_t + \left[\frac{s}{(1+n)(1+g)} \right] f(k_t).$$

(b) We need to sketch k_{t+1} as a function of k_t .

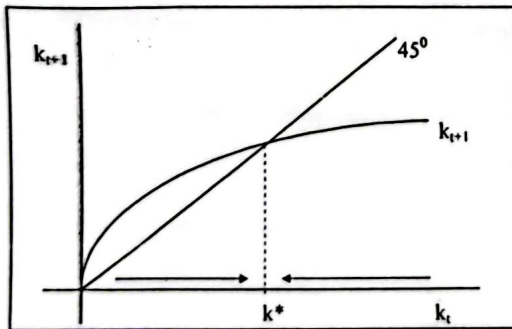
Note that

$$(4) \frac{\partial k_{t+1}}{\partial k_t} = \frac{1-\delta}{(1+n)(1+g)} + \left[\frac{s}{(1+n)(1+g)} \right] f'(k_t) > 0, \quad \text{and} \quad (5) \frac{\partial^2 k_{t+1}}{\partial k_t^2} = \frac{sf''(k_t)}{(1+n)(1+g)} < 0.$$

The Inada conditions are given by

$$(6) \lim_{k \rightarrow 0} \frac{\partial k_{t+1}}{\partial k_t} = \infty, \quad \text{and} \quad (7) \lim_{k \rightarrow \infty} \frac{\partial k_{t+1}}{\partial k_t} = \frac{1-\delta}{(1+n)(1+g)} < 1.$$

Thus the function eventually has a slope of less than one and will therefore cross the 45 degree line at some point. Also, the function is well-behaved and will cross the 45 degree line only once.



As long as k starts out at some value other than 0, the economy will converge to k^* . For example, if k starts out below k^* , we see that k_{t+1} will be greater than k_t and the economy will move toward k^* . Similarly, if k starts out above k^* , we see that k_{t+1} will be below k_t and again the economy will move toward k^* . At k^* , $y^* = f(k^*)$ is also constant and we have a balanced growth path.

(c) On a balanced growth path, $k_{t+1} = k_t \equiv k^*$ and thus from equation (3) we have

$$(8) k^* = \left[\frac{1-\delta}{(1+n)(1+g)} \right] k^* + \left[\frac{s}{(1+n)(1+g)} \right] f(k^*),$$

which simplifies to

$$(9) k^* \left[\frac{1+n+g+ng-1+\delta}{(1+n)(1+g)} \right] = \left[\frac{s}{(1+n)(1+g)} \right] f(k^*).$$

Thus on a balanced growth path:

$$(10) k^*(n+g+ng+\delta) = sf(k^*).$$

Rearranging equation (10) to get an expression for s on the balanced growth path yields

$$(11) s = (n+g+ng+\delta)k^*/f(k^*).$$

Consumption per unit of effective labor on the balanced growth path is given by

$$(12) c^* = (1-s)f(k^*).$$

Substitute equation (11) into equation (12):

$$(13) c^* = \left[1 - \frac{(n+g+ng+\delta)k^*}{f(k^*)} \right] f(k^*) = \left[\frac{f(k^*) - k^*(n+g+ng+\delta)}{f(k^*)} \right] f(k^*).$$

Canceling the $f(k^*)$ yields

$$(14) c^* = f(k^*) - (n+g+ng+\delta)k^*.$$

To get an expression for the $f'(k^*)$ that maximizes consumption per unit of effective labor on the balanced growth path, we need to maximize c^* with respect to k^* . The first-order condition is given by

$$(15) \partial c^* / \partial k^* = f'(k^*) - (n+g+ng+\delta) = 0.$$

Thus the golden-rule capital stock is defined implicitly by

$$(16) f'(k_{GR}) = (n+g+ng+\delta).$$

(d) (i) Substitute a Cobb-Douglas production function, $f(k_t) = k_t^\alpha$, into equation (3):

$$(17) k_{t+1} = \left[\frac{1-\delta}{(1+n)(1+g)} \right] k_t + \left[\frac{s}{(1+n)(1+g)} \right] k_t^\alpha.$$

(d) (ii) On a balanced growth path, $k_{t+1} = k_t \equiv k^*$. Thus from equation (17):

$$(18) k^* = \left[\frac{1-\delta}{(1+n)(1+g)} \right] k^* + \left[\frac{s}{(1+n)(1+g)} \right] k^{*\alpha}.$$

Simplifying yields

$$(19) \left[\frac{(1+n)(1+g) - (1-\delta)}{(1+n)(1+g)} \right] k^* = \left[\frac{s}{(1+n)(1+g)} \right] k^{*\alpha},$$

or

$$(20) k^{*1-\alpha} = s / (n+g+ng+\delta).$$

Thus, finally we have

$$(21) k^* = [s / (n+g+ng+\delta)]^{1/(1-\alpha)}.$$

(d) (iii) Using equation (17):

$$(22) \left. \frac{dk_{t+1}}{dk_t} \right|_{k_t=k^*} = \frac{1-\delta}{(1+n)(1+g)} + \frac{\alpha s}{(1+n)(1+g)} k^{*\alpha-1}.$$

Substituting the balanced-growth-path value of k^* – equation (21) – into equation (22) yields

$$(23) \left. \frac{dk_{t+1}}{dk_t} \right|_{k_t=k^*} = \frac{1-\delta}{(1+n)(1+g)} + \frac{\alpha s}{(1+n)(1+g)} \left[\frac{(n+g+ng+\delta)}{s} \right].$$

It will be useful to write $(n+g+ng+\delta)$ as $(1+n)(1+g) - (1-\delta)$:

$$(24) \left. \frac{dk_{t+1}}{dk_t} \right|_{k_t=k^*} = \frac{(1-\delta) + \alpha[(1+n)(1+g) - (1-\delta)]}{(1+n)(1+g)}.$$

Simplifying further yields

$$(25) \left. \frac{dk_{t+1}}{dk_t} \right|_{k_t=k^*} = \alpha + \frac{(1-\delta)(1-\alpha)}{(1+n)(1+g)}.$$

Replacing equation (17) by its first-order Taylor approximation around $k = k^*$ therefore gives us

$$(26) k_{t+1} \cong k^* + \left[\alpha + (1-\delta)(1-\alpha)/(1+n)(1+g) \right] [k_t - k^*].$$

Since we can write this simply as

$$(27) k_{t+1} - k^* \cong \left[\alpha + (1-\delta)(1-\alpha)/(1+n)(1+g) \right] [k_t - k^*],$$

equation (26) implies

$$(28) k_t - k^* \cong \left[\alpha + (1-\delta)(1-\alpha)/(1+n)(1+g) \right]^t [k_0 - k^*].$$

Thus the economy moves fraction $1 - [\alpha + (1-\delta)(1-\alpha)/(1+n)(1+g)]$ of the way to the balanced growth path each period. Some simple algebra simplifies the expression for this rate of convergence to $(1-\alpha)(n+g+ng+\delta)/(1+n)(1+g)$. With $\alpha = 1/3$, $n = 1\%$, $g = 2\%$ and $\delta = 3\%$, this yields a rate of convergence of about 3.9%. This is slower than the rate of convergence found in the continuous-time Solow model.

Problem 2.16

(a) The individual's optimization problem is not affected by the depreciation which means that $r_t = f'(k_t) - \delta$. The household's problem is still to maximize utility as given by

$$(1) U_t = \frac{C_{1,t}^{1-\theta}}{1-\theta} + \frac{1}{1+\rho} \frac{C_{2,t+1}^{1-\theta}}{1-\theta},$$

subject to the budget constraint

$$(2) C_{1,t} + \frac{1}{1+r_{t+1}} C_{2,t+1} = A_t w_t.$$

As in the text, with no depreciation, the fraction of income saved, $s(r_{t+1}) \equiv (1 - C_{1,t})/A_t w_t$, is given by

$$(3) s(r_{t+1}) = \frac{1}{1 + (1+\rho)^{1/\theta} (1+r_{t+1})^{(\theta-1)/\theta}}.$$

Thus the way in which the fraction of income saved depends on the real interest rate, r_{t+1} , is unchanged.

The only difference is that the real interest rate itself is now $f'(k_{t+1}) - \delta$, rather than just $f'(k_{t+1})$.

The capital stock in period $t+1$ equals the amount saved by young individuals in period t . Thus

$$(4) K_{t+1} = S_t L_t,$$

where S_t is the amount of saving done by a young person in period t . Note that $S_t \equiv s(r_{t+1}) A_t w_t$; the amount of saving done is equal to the fraction of income saved times the amount of income. Thus equation (4) can be rewritten as

$$(5) K_{t+1} = L_t s(r_{t+1}) A_t w_t.$$

To get this into units of time $t+1$ effective labor, divide both sides of equation (5) by $A_{t+1} L_{t+1}$:

$$(6) \frac{K_{t+1}}{A_{t+1} L_{t+1}} = \frac{A_t L_t}{A_{t+1} L_{t+1}} [s(r_{t+1}) w_t].$$

Since $A_{t+1} = (1+g)A_t$, we have $A_t/A_{t+1} = 1/(1+g)$. Similarly, $L_t/L_{t+1} = 1/(1+n)$. In addition, $K_{t+1}/A_{t+1} L_{t+1} \equiv k_{t+1}$. Thus

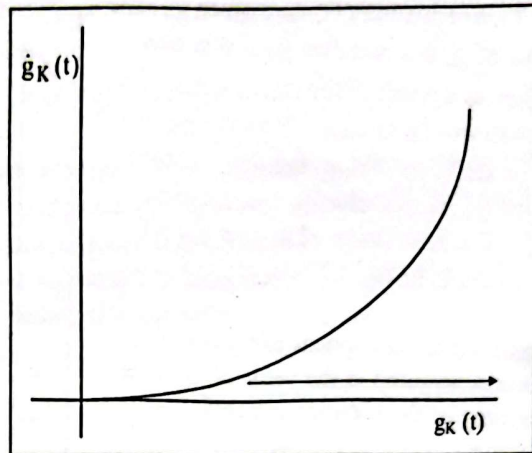
$$(7) k_{t+1} = \frac{1}{(1+n)(1+g)} [s(r_{t+1}) w_t].$$

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Equation (4) is plotted at right. With $\theta > 1$, $g_{K,i}$ will be always increasing. The initial value of $g_{K,i}$ is determined by the initial capital stock and the saving rate; see equation (2).

Since both economies have the same $K(0)$ but one has a higher saving rate, then from equation (2), the economy with the higher s will have the higher initial $g_{K,i}(0)$.

From equation (3), the growth rate of $g_{K,i}$ is increasing in $g_{K,i}$. Thus the growth rate of the capital stock in the high-saving economy will always exceed the growth rate of the capital stock in the low-saving economy. That is, we have $g_{K,1}(t) > g_{K,2}(t)$ for all $t \geq 0$. In fact, the gap between the two growth rates will be increasing over time.



More formally, using the production function, we can write the ratio of output in the high-saving country, country 1, to output in the low-saving country, country 2, as

$$(5) \quad Y_1(t)/Y_2(t) = [K_1(t)/K_2(t)]^\theta.$$

Taking the time derivative of the log of equation (5) yields an expression for the growth rate of the ratio of output in the high-saving economy to output in the low-saving economy:

$$(6) \quad \frac{[Y_1(t)/Y_2(t)]}{[Y_1(t)/Y_2(t)]} = \theta \left[\frac{\dot{K}_1(t)}{K_1(t)} - \frac{\dot{K}_2(t)}{K_2(t)} \right] = \theta [g_{K,1}(t) - g_{K,2}(t)] > 0.$$

As explained above, $g_{K,1}(t)$ will exceed $g_{K,2}(t)$ for all $t \geq 0$. In fact, the gap between the two will be increasing over time. Thus the growth rate of the output ratio will be positive and increasing over time. That is, the ratio of output in the high-saving economy to output in the low-saving economy will be continually rising, and rising at an increasing rate.

Problem 3.3

The equations of the $\dot{g}_K = 0$ and $\dot{g}_A = 0$ lines are given by

$$(1) \quad \dot{g}_K = 0 \Rightarrow g_K = g_A + n,$$

and

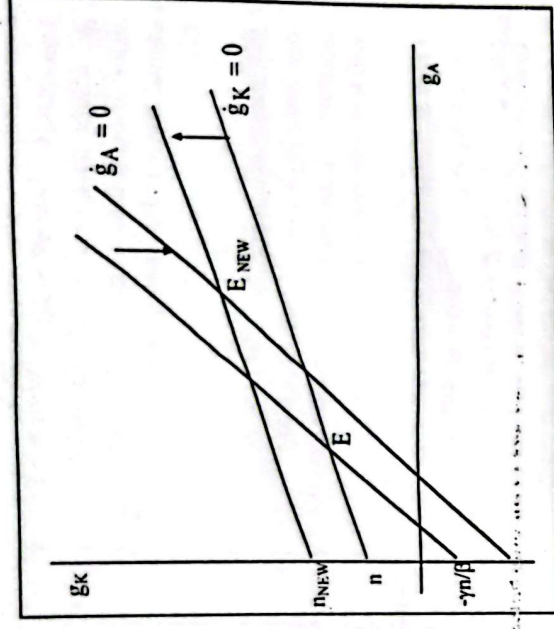
$$(2) \quad \dot{g}_A = 0 \Rightarrow g_K = \frac{(1-\theta)g_A - \gamma n}{\beta}.$$

The expressions for the growth rates of capital and knowledge are

$$(3) \quad g_K(t) = c_K [A(t)L(t)/K(t)]^{1-\alpha} \quad c_K \equiv s(1-a_K)^\alpha (1-a_L)^{1-\alpha}$$

$$(4) \quad g_A(t) = c_A K(t)^\beta L(t)^\gamma A(t)^{\theta-1} \quad c_A \equiv B a_K^\beta a_L^\gamma.$$

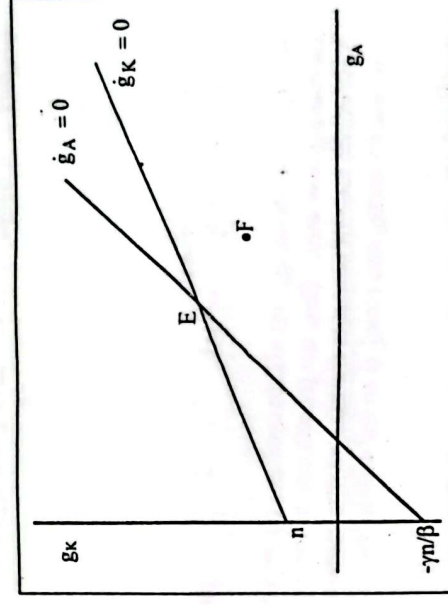
(a) From equation (1), for a given g_A , the value of g_K that satisfies $\dot{g}_K = 0$ is now higher as a result of the rise in population growth from n to n_{NEW} . Thus the $\dot{g}_K = 0$ locus shifts up. From equation (2), for a given g_A , the value of g_K that satisfies $\dot{g}_A = 0$ is now lower. Thus the $\dot{g}_A = 0$ locus shifts down.



Since n does not appear in equation (3), there is no jump in the value of g_K at the moment of the increase in population growth.

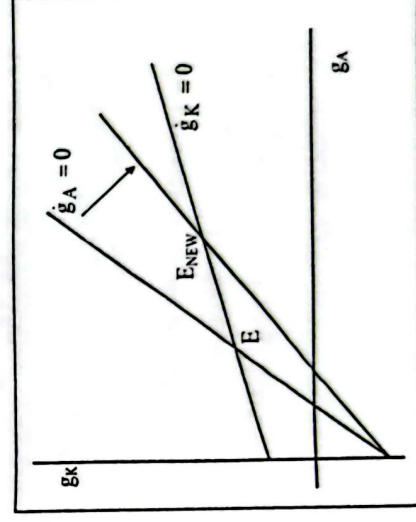
Similarly, since n does not appear in equation (4), there is no jump in the value of g_A at the moment of the rise in population growth.

(b) Note that a_K does not appear in equation (1), the $\dot{g}_K = 0$ line, or in equation (2), the $\dot{g}_A = 0$ line. Thus neither the $\dot{g}_K = 0$ nor the $\dot{g}_A = 0$ line shifts as a result of the increase in the fraction of the capital stock used in the knowledge sector from a_K to a_K^{NEW} .



From equation (3), the rise in a_K causes the growth rate of capital, g_K , to jump down. From equation (4), the growth rate of knowledge, g_A , jumps up at the instant of the rise in a_K . Thus the economy moves to a point such as F in the figure.

(c) Since θ does not appear in equation (1), there is no shift of the $\dot{g}_K = 0$ locus as a result of the rise in θ , the coefficient on knowledge in the knowledge production function. From equation (2), the $\dot{g}_A = 0$ locus has slope $(1 - \theta)/\beta$ and therefore becomes flatter after the rise in θ . See the figure.



Since θ does not appear in equation (3), the growth rate of capital, g_K , does not jump at the time of the rise in θ . θ does appear in equation (4) and thus we need to determine the effect that the rise in θ has on the growth rate of knowledge. It turns out

that g_A may jump up, jump down or stay the same at the instant of the change in θ . Taking the log of both sides of equation (4) gives us

$$(5) \ln g_A(t) = \ln c_A + \beta \ln K(t) + \gamma \ln L(t) + (\theta - 1) \ln A(t).$$

Taking the derivative of both sides of equation (5) with respect to θ yields

$$(6) \partial \ln g_A(t) / \partial \theta = \ln A(t).$$

So if $A(t)$ is less than one, so that $\ln A(t) < 0$, the growth rate of knowledge jumps down at the instant of the rise in θ . However, if $A(t)$ is greater than one, so that $\ln A(t) > 0$, the growth rate of knowledge jumps up at the instant of the rise in θ . Finally, if $A(t)$ is equal to one at the time of the change in θ , there is no initial jump in g_A . This means the dynamics of the adjustment to E_{NEW} may differ depending on the value of g_A at the time of the change in θ , but the end result is the same.

Problem 3.4

The equations of the $\dot{g}_K = 0$ and $\dot{g}_A = 0$ loci are

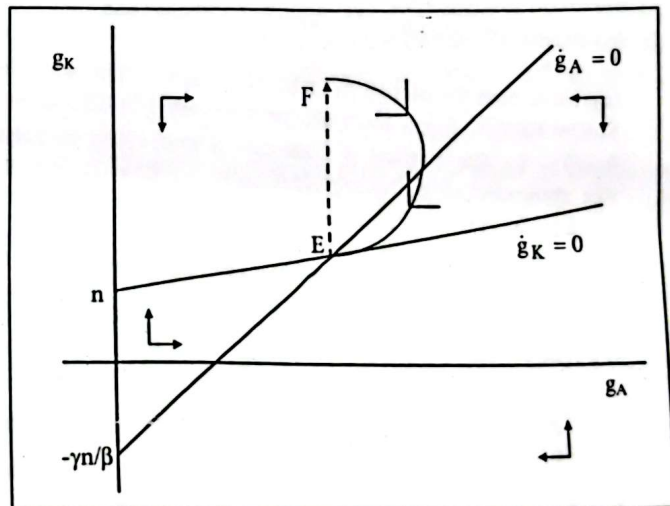
$$(1) \dot{g}_K = 0 \Rightarrow g_K = g_A + n, \text{ and } (2) \dot{g}_A = 0 \Rightarrow g_K = \frac{(1-\theta)g_A - \gamma n}{\beta}.$$

The equations defining the growth rates of capital and knowledge at any point in time are

$$(3) g_K(t) = c_K [A(t)L(t)/K(t)]^{1-\alpha} \quad c_K \equiv s(1-a_K)^\alpha (1-a_L)^{1-\alpha}$$

$$(4) g_A(t) = c_A K(t)^\beta L(t)^\gamma A(t)^{\theta-1} \quad c_A \equiv B a_K^\beta a_L^\gamma.$$

(a) Since the saving rate, s , does not appear in equations (1) or (2), neither the $\dot{g}_K = 0$ nor the $\dot{g}_A = 0$ locus shifts when s increases. From equation (4), the growth rate of knowledge, g_A , does not change at the moment that s increases. However, from equation (3), a rise in s causes an upward jump in the growth rate of capital, g_K . In the figure, the economy jumps from its balanced growth path at E to a point such as F at the moment that s increases.



(b) At point F , the economy is above the $\dot{g}_A = 0$ locus and thus g_A is rising.

Due to the increase in s , the growth rate of capital is higher than it would have been – the amount of capital going into the production of knowledge is higher than it would have been – and so the growth rate of knowledge begins to rise above what it would have been. Also at point F , the economy is above the $\dot{g}_K = 0$ locus and so g_K is falling. The economy drifts to the southeast and eventually crosses the $\dot{g}_A = 0$ locus at which point g_A begins to fall as well. Since there are decreasing returns to capital and knowledge in the production of new knowledge – $\theta + \beta < 1$ – the increase in s does not have a permanent effect on the growth rates of K and A . The economy eventually returns to point E .

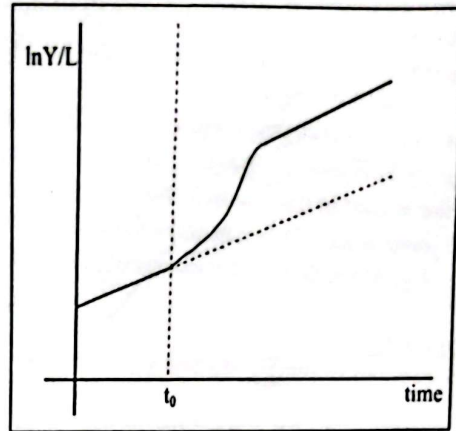
The production function is given by

$$(5) Y(t) = [(1-a_K)K(t)]^\alpha [A(t)(1-a_L)L(t)]^{1-\alpha}.$$

Taking the time derivative of the log of equation (5) will yield the growth rate of total output:

$$(6) \quad \frac{\dot{Y}(t)}{Y(t)} = \alpha g_K(t) + (1-\alpha)[g_A(t) + n].$$

On the initial balanced growth path, from equation (1), $g_K^* = g_A^* + n$. From equation (6), this means that total output is also growing at rate $g_K^* = g_A^* + n$ on the initial balanced growth path. Thus output per person, $Y(t)/L(t)$, is initially growing at rate g_A^* . During the transition period, both g_K and g_A are higher than on the balanced growth path and so output per worker must be growing at a rate greater than its balanced-growth-path value of g_A^* . Whether the growth rate of output per worker is rising or



falling will depend, among other things, on the value of α since there is a period of time when g_K is falling and g_A is rising. The figure shows the growth rate of output per worker initially rising and then falling, but the important point is that during the entire transition, the growth rate itself is higher than its balanced-growth-path value of g_A^* . In the end, once the economy returns to point E, output per worker is again growing at rate g_A^* , which has not changed.

(c) Note that the effects of an increase in s in this model are qualitatively similar to the effects in the Solow model. Since there are net decreasing returns to the produced factors of production here – $\theta + \beta < 1$ – the increase in s has only a level effect on output per worker. The path of output per worker lies above the path it would have taken but there is no permanent effect on the growth rate of output per worker, which on the balanced growth path is equal to the growth rate of knowledge. This is the same effect that a rise in s has in the Solow model in which there are diminishing returns to the produced factor, capital. Quantitatively, the effect is larger than in the Solow model (for a given set of parameters). This is due to the fact that, here, A rises above the path it would have taken whereas that is not true in the Solow model.

Problem 3.5

(a) From equations (3.14) and (3.16) in the text, the growth rates of capital and knowledge are given by

$$(1) \quad g_K(t) \equiv \dot{K}(t)/K(t) = c_K [A(t)L(t)/K(t)]^{1-\alpha}, \quad \text{where } c_K \equiv s[1 - a_K]^\alpha [1 - a_L]^{1-\alpha}, \text{ and}$$

$$(2) \quad g_A(t) \equiv \dot{A}(t)/A(t) = c_A K(t)^\beta L(t)^\gamma A(t)^{\theta-1}, \quad \text{where } c_A \equiv B a_K^\beta a_L^\gamma.$$

With the assumptions of $\beta + \theta = 1$ and $n = 0$, these equations simplify to

$$(3) \quad g_K(t) = [c_K L^{1-\alpha}] [A(t)/K(t)]^{1-\alpha}, \quad \text{and} \quad (4) \quad g_A(t) = [c_A L^\gamma] [K(t)/A(t)]^\beta.$$

Thus given the parameters of the model and the population (which is constant), the ratio A/K determines both growth rates. The two growth rates, g_K and g_A , will be equal when

$$(5) \quad [c_K L^{1-\alpha}] [A(t)/K(t)]^{1-\alpha} = [c_A L^\gamma] [K(t)/A(t)]^\beta, \quad \text{or when}$$

$$(6) \quad [A(t)/K(t)]^{1-\alpha+\beta} = [c_A/c_K] L^{\gamma-(1-\alpha)}.$$

Thus the value of A/K that yields equal growth rates of capital and knowledge is given by

$$(7) \quad A(t)/K(t) = \left[(c_A/c_K) L^{\gamma-(1-\alpha)} \right]^{1/(1-\alpha+\beta)}.$$

(b) In order to find the growth rate of A and K when $g_K = g_A \equiv g^*$, substitute equation (7) into (3):

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$$(8) \ g^* = [c_K L^{1-\alpha}] [(c_A/c_K) L^{\gamma-(1-\alpha)}]^{(1-\alpha)/(1-\alpha+\beta)}$$

Simplifying the exponents yields

$$(9) \ g^* = [c_K^{(1-\alpha+\beta)-(1-\alpha)} c_A^{1-\alpha} L^{(1-\alpha)+\gamma(1-\alpha)-(1-\alpha)^2}]^{1/(1-\alpha+\beta)},$$

or simply

$$(10) \ g^* = [c_K^\beta c_A^{1-\alpha} L^{(1-\alpha)(\gamma+\alpha)}]^{1/(1-\alpha+\beta)}$$

(c) In order to see the way in which an increase in s affects the long-run growth rate of the economy, substitute the definitions of c_K and c_A into equation (10):

$$(11) \ g^* = [s^\beta (1-a_K)^\alpha \beta (1-a_L)^{(1-\alpha)\beta} B^{1-\alpha} a_K^\beta (1-a_L)^\alpha a_L^\gamma L^{(1-\alpha)(\gamma+\alpha)}]^{1/(1-\alpha+\beta)}$$

Taking the natural logarithm of both sides of equation (11) gives us

$$(12) \ \ln g^* = [1/(1-\alpha+\beta)] \{ \beta \ln s + (1-\alpha)(\gamma+\alpha) \ln L + (1-\alpha) \ln B + \beta[\alpha \ln(1-a_K) + (1-\alpha) \ln a_K] + (1-\alpha)[\beta \ln(1-a_L) + \gamma \ln a_L] \}$$

Using equation (12), the elasticity of the long-run growth rate of the economy with respect to the saving rate is

$$(13) \ \partial \ln g^* / \partial \ln s = \beta / (1-\alpha+\beta) > 0.$$

Thus an increase in the saving rate increases the long-run growth rate of the economy. This is essentially because it increases the resources devoted to physical capital accumulation and in this model, we have constant returns to the produced factors of production.

(d) We can maximize $\ln g^*$ with respect to a_K to determine the fraction of the capital stock that should be employed in the R&D sector in order to maximize the long-run growth rate of the economy.

The first-order condition is

$$(14) \ \frac{\partial \ln g^*}{\partial a_K} = \frac{\beta}{(1-\alpha+\beta)} \left[\frac{-\alpha}{(1-a_K)} + \frac{(1-\alpha)}{a_K} \right] = 0.$$

Solving for the optimal a_K^* yields

$$(15) \ \alpha/(1-a_K) = (1-\alpha)/a_K.$$

Simplifying gives us

$$(16) \ \alpha a_K = 1 - a_K + \alpha a_K - \alpha,$$

and thus

$$(17) \ a_K^* = (1-\alpha).$$

Thus the optimal fraction of the capital stock to employ in the R&D sector is equal to effective labor's share in the production of output. Note that β , capital's share in the production function for new knowledge, does not affect the optimal allocation of capital to the R&D sector. The reason for this is that an increase in β has two effects. It makes capital more important in the R&D sector, thereby tending to raise the a_K that maximizes g^* . A rise in β also makes the production of new capital more valuable, and new capital is produced when there is more output to be saved and invested. This tends to lower the a_K that maximizes g^* since it implies that more resources should be devoted to the production of output rather than knowledge. In the case we are considering, these two effects exactly cancel each other out.